

Day 4: Input Handling

- Scanner class
- BufferedReader
- Command line arguments

Programs:

- User input calculator
- Student details

Real World Story First

Imagine...

Mee friend ni adugutunnaru.

"Nee peru enti?"

Friend:

"Alex"

Meeru adugutunnaru

"Age enti?"

Friend:

"20"

Meeru adugutunnaru

"College enti?"

Friend:

"Institute"

Ikkada

You → Program

Friend → User

User answer ivvadam ante **Input**

Program answer ivvadam ante **Output**

Day 1 lo

```
System.out.println("Hello");
```

Output icham.

Today

User daggara nundi information ela tiskovalo nerchukuntam.

What is Input?

Definition:

Input ante

Program execute avuthunnappudu user nundi data receive cheyyadam.

Examples

ATM

Enter PIN:

Calculator

Enter first number:

Instagram

Enter Username

Enter Password

Google

Search Here...

Ivanni Input.

Java lo Input ela tiskoali?

Java lo mainly 3 methods unnayi.

1.Scanner Class ☆☆☆☆☆ (Most Used)

2.BufferedReader ☆☆☆

3.Command Line Arguments ☆☆☆

Interview lo kuda adugutaru.

1.Scanner Class

Idi Java lo most famous input class.

Almost 95% beginners Scanner use chestaru.

Scanner ekkada untundi?

Scanner class

java.util package

lo untundi.

Kabatti import cheyyali.

```
import java.util.Scanner;
```

Scanner syntax

```
import java.util.Scanner;
```

```
Scanner sc = new Scanner(System.in);
```

Idi chala important line.

By heart kakunda meaning ardham chesukondi.

Let's divide.

Scanner

Class name

sc

Object name

Object peru edaina pettachu.

Scanner input

Scanner abc

Scanner myScanner

Next

new Scanner(System.in)

Meaning

Create new Scanner object.

What is System.in ?

Already

System.out.println()

use chesam.

out

means Output.

System.out

Keyboard kaadu.

Adi Screen.

Mari

System.in

Ante

Keyboard input.

Diagram

Keyboard

↓

System.in

↓

Scanner

↓

Variable

↓

Program

Scanner Flow

User



Keyboard



System.in



Scanner



Variable



Program

Example

Enter Age

20

Flow

20



Keyboard



Scanner

↓

age variable

↓

Print

First Scanner Program

```
import java.util.Scanner;
```

```
public class Main
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter your name:");
```

```
        String name = sc.nextLine();
```

```
        System.out.println("Welcome " + name);
```

```
    }
```

```
}
```

Output

Enter your name

Alex

Welcome Alex

Scanner Methods

Scanner different data types read cheyyagaladu.

Most Important Methods

`next()`

`nextLine()`

`nextInt()`

`nextDouble()`

`nextFloat()`

`nextLong()`

`nextBoolean()`

`next().charAt(0)`

Okkokati deep ga chuddam.

`next()`

One word matrame read chestundi.

Example

Input

Ram Ayodhya

Code

```
String name = sc.next();
```

Output

Ram

Ayodhya ignore chestundi.

Because

next()

Stops at Space.

Diagram

Ram Ayodhya

^

Stops Here

nextLine()

Entire sentence read chestundi.

Example

Ram Ayodhya

Code

```
String name=sc.nextLine();
```

Output

Ram Ayodhya

Entire line.

Difference

next()

↓

One Word

`nextLine()`

↓

Entire Line

Interview Question.

`nextInt()`

Integer read chestundi.

Example

```
int age=sc.nextInt();
```

Input

20

Output

20

`nextDouble()`

Decimal read chestundi.

```
double salary=sc.nextDouble();
```

Input

25000.75

`nextFloat()`

```
float marks=sc.nextFloat();
```

Input

90.5

nextLong()

long mobile=sc.nextLong();

nextBoolean()

Input

true

or

false

boolean value=sc.nextBoolean();

Character Input

Scanner lo

nextChar()

ledu.

So

char ch=sc.next().charAt(0);

Explanation

next()

↓

Reads String

↓

charAt(0)

↓

First Character

Input

Ram

Output

R

charAt()

Example

JAVA

Index

J A V A

0 1 2 3

charAt(0)

↓

J

charAt(2)

↓

V

Scanner Complete Example

```
import java.util.Scanner;
```

```
public class Main
{
    public static void main(String args[])
    {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Name: ");
        String name=sc.nextLine();

        System.out.print("Enter Age: ");
        int age=sc.nextInt();

        System.out.print("Enter Height: ");
        float height=sc.nextFloat();

        System.out.print("Enter Gender(M/F): ");
        char gender=sc.next().charAt(0);

        System.out.println(name);
        System.out.println(age);
        System.out.println(height);
        System.out.println(gender);

    }
}
```

Why close Scanner?

After work complete ayyaka

```
sc.close();
```

use chestaru.

Meaning

Scanner resource ni close chestundi.

Like

Tap open chesi water use chesi close chestam kada.

Alage.

Problem with nextLine()

Idi interview lo famous.

Example

```
int age=sc.nextInt();
```

```
String name=sc.nextLine();
```

Input

20

Ram

Output

age=20

name=

Empty.

Why?

Because

nextInt()

Number matrame read chestundi.

Enter key

(\n)

buffer lo untundi.

nextLine()

A newline ni read chesi empty string return chestundi.

Solution

```
int age=sc.nextInt();
```

```
sc.nextLine();
```

```
String name=sc.nextLine();
```

Middle lo

```
sc.nextLine();
```

newline remove chestundi.

What is Buffer?

Imagine

Keyboard nundi vachina characters temporary ga store ayye waiting room ni **Buffer** antaru.

Example:

Meeru type chesaru:

20↵

Memory lo temporary ga ila untundi:

2 0 \n

nextInt():

- 20 ni read chestundi

- \n (Enter) ni vadilesi buffer lo unchestundi

nextLine() vachinappudu:

- Buffer lo unna \n ni read chesi empty line return chestundi.

Kabatti extra sc.nextLine(); use chestam.

2. BufferedReader

Scanner kanna mundu

BufferedReader use chesevallu.

Still large applications lo use chestaru.

Package

java.io

Import

```
import java.io.*;
```

Syntax

```
BufferedReader br =
```

```
new BufferedReader(
```

```
new InputStreamReader(System.in));
```

Looks difficult.

But divide chesithe easy.

Keyboard

↓

System.in

↓

InputStreamReader

↓

BufferedReader

↓

Program

Read String

```
String name=br.readLine();
```

Input

ram

Output

ram

Read Integer

Problem

BufferedReader

Always

Returns String.

So convert cheyyali.

```
int age=Integer.parseInt(br.readLine());
```

Flow

Input

↓

String

↓

`parseInt()`

↓

Integer

Decimal

`double d=`

`Double.parseDouble(br.readLine());`

BufferedReader Example

`import java.io.*;`

`public class Main`

`{`

`public static void main(String args[]) throws Exception`

`{`

`BufferedReader br =`

`new BufferedReader(`

`new InputStreamReader(System.in));`

`System.out.print("Enter Name: ");`

`String name=br.readLine();`

`System.out.print("Enter Age: ");`

```

        int age=Integer.parseInt(br.readLine());

        System.out.println(name);
        System.out.println(age);
    }
}

```

Why throws Exception?

readLine() file/device nundi data read chestundi. Reading process lo error ravachu (IOException).

Beginner level lo easiest way:

```
public static void main(String[] args) throws Exception
```

Advanced ga try-catch nerchukuntaru later.

Scanner vs BufferedReader

Feature	Scanner	BufferedReader
Easy for beginners	✓	✗
Reads different data types directly	✓	✗ (Need parsing)
Speed	Slower	Faster
Methods	Many (nextInt, nextDouble)	Mostly readLine()
Best for	Beginners, Competitive Coding	Large input, High Performance

Which One Should I Learn?

Beginner

↓

Scanner

Competitive Programming

↓

Scanner or Fast Input classes

Large Applications

↓

BufferedReader

3. Command Line Arguments

Idi chala interesting topic.

Program run chestunnappude values pampadam.

Keyboard nundi kaadu.

Command nundi.

Example

```
public static void main(String args[])
```

Ikkada

args

already chusaru.

Adi waste kaadu.

Adi

Command Line Arguments.

Example

Command Prompt lo:

```
java Main ram 20 CSE
```

Ila run chesthe

```
args[0]
```

↓

Harika

args[1]

↓

20

args[2]

↓

CSE

Diagram

Command

↓

java Main ram 20 CSE

↓

args[]

↓

Program

Example

```
public class Main
```

```
{  
    public static void main(String args[])  
    {  
  
        System.out.println(args[0]);  
  
        System.out.println(args[1]);  
  
    }  
}
```

Run

java Main ram Ayodhya

Output

raa

Ayodhya

Convert Number

int age=

Integer.parseInt(args[0]);

If No Arguments?

Example

System.out.println(args[0]);

Run

java Main

Output

Exception

Reason:

Array empty.

Kabatti check cheyyali.

```
if(args.length > 0){  
    System.out.println(args[0]);  
}else{  
    System.out.println("No arguments provided");  
}
```

Scanner vs Command Line Arguments

Scanner

User execution time lo input istadu Program start ayye mundu arguments istaru

Interactive

Keyboard use

Most common

Command Line Arguments

Non-interactive

Command use

Special cases, automation

Program 1: User Input Calculator

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter first number: ");  
        double a = sc.nextDouble();  
  
        System.out.print("Enter second number: ");  
        double b = sc.nextDouble();
```

```

System.out.print("Enter operator (+, -, *, /): ");

char op = sc.next().charAt(0);

switch (op) {
    case '+':
        System.out.println("Result = " + (a + b));
        break;
    case '-':
        System.out.println("Result = " + (a - b));
        break;
    case '*':
        System.out.println("Result = " + (a * b));
        break;
    case '/':
        if (b != 0) {
            System.out.println("Result = " + (a / b));
        } else {
            System.out.println("Division by zero is not allowed.");
        }
        break;
    default:
        System.out.println("Invalid operator.");
}

sc.close();
}
}

```

Dry Run

Input:

Enter first number: 10

Enter second number: 5

Enter operator (+, -, *, /): *

Execution:

- a = 10
- b = 5
- op = '*'
- switch * case execute avutundi.
- Output = 10 * 5 = 50

Output:

Result = 50.0

Program 2: Student Details

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter Student Name: ");  
        String name = sc.nextLine();  
  
        System.out.print("Enter Roll Number: ");  
        int roll = sc.nextInt();  
  
        System.out.print("Enter Marks: ");  
        float marks = sc.nextFloat();  
    }  
}
```

```
sc.nextLine(); // Consume leftover newline

System.out.print("Enter Course: ");
String course = sc.nextLine();

System.out.println("\n----- Student Details -----");
System.out.println("Name   : " + name);
System.out.println("Roll   : " + roll);
System.out.println("Marks  : " + marks);
System.out.println("Course : " + course);

sc.close();
}
}
```

Dry Run

Input:

Enter Student Name: Ram Ayodhya

Enter Roll Number: 101

Enter Marks: 92.5

Enter Course: CSE

Variables:

- name = "Ram Ayodhya"
- roll = 101
- marks = 92.5
- course = "CSE"

Output:

----- Student Details -----

Name : Ram Ayodhya

Roll : 101

Marks : 92.5

Course : CSE

Interview Questions (Must Prepare)

1. What is Input Handling in Java?

Answer:

Input Handling ante **user nundi data ni program lo receive cheyyadam**.

Example:

- Name
- Age
- Marks
- Salary

User keyboard dwara data enter chestadu, Java program aa data ni process chestundi.

Example:

```
Scanner sc = new Scanner(System.in);
```

```
int age = sc.nextInt();
```

2. What is Scanner Class?

Answer:

Scanner anedi **java.util package** lo unna predefined class.

Idi **keyboard nundi input** tiskovadaniki use chestam.

Scanner different data types ni direct ga read cheyyagaladu.

Examples:

- nextInt()
- nextDouble()
- next()
- nextLine()

- nextBoolean()

Import:

```
import java.util.Scanner;
```

3. Why do we write new Scanner(System.in)?

Answer:

new Scanner(System.in) anedi Scanner object create cheyyadaniki use chestam.

- new → new object create chestundi.
- Scanner() → Scanner constructor.
- System.in → Keyboard input stream.

Without creating object, Scanner methods use cheyyalem.

Example:

```
Scanner sc = new Scanner(System.in);
```

4. What is System.in?

Answer:

System.in anedi **standard input stream**.

Adi keyboard nundi data receive chestundi.

Flow:

Keyboard

↓

System.in

↓

Scanner

↓

Program

5. Difference between next() and nextLine().

next()

nextLine()

Reads only one word Reads complete line

Stops at space Reads until Enter key

Cannot read spaces Can read spaces

Example Input:

Harika Meka

```
sc.next();
```

Output:

Harika

```
sc.nextLine();
```

Output:

Harika Meka

6. Why do we use `sc.close()`?

Answer:

`sc.close()` Scanner object ni close cheyyadaniki use chestam.

Benefits:

- Releases system resources.
- Prevents memory/resource leaks.
- Good programming practice.

Example:

```
sc.close();
```

7. Why does `nextLine()` sometimes return an empty string after `nextInt()`?

Answer:

`nextInt()` number matrame read chestundi.

User Enter press chesina newline (`\n`) buffer lo untundi.

`nextLine()` aa newline ni read chesi empty string return chestundi.

Solution:

```
int age = sc.nextInt();
```

```
sc.nextLine();
```

```
String name = sc.nextLine();
```

8. What is a Buffer?

Answer:

Buffer ante **temporary memory area**.

Keyboard nundi vachina data temporary ga buffer lo store avutundi.

Example:

Input:

20↵

Buffer:

2 0 \n

nextInt() → 20 read chestundi.

\n buffer lo untundi.

9. What is BufferedReader?

Answer:

BufferedReader anedi **java.io package** lo unna predefined class.

Idi keyboard nundi input read cheyyadaniki use chestam.

Scanner kanna faster.

Import:

```
import java.io.*;
```

10. Why is BufferedReader faster than Scanner?

Answer:

BufferedReader complete line ni okesari read chestundi.

Scanner input ni parse chestu untundi kabatti konchem slow.

Therefore:

- BufferedReader → Faster
 - Scanner → Easier
-

11. Why do we use Integer.parseInt() with BufferedReader?

Answer:

BufferedReader readLine() method always String return chestundi.

Integer kavali ante String ni int ga convert cheyyali.

Example:

```
int age = Integer.parseInt(br.readLine());
```

Flow:

String

↓

parseInt()

↓

int

12. What are Command Line Arguments?

Answer:

Command Line Arguments ante program run chese time lo command prompt nundi values pass cheyyadam.

Keyboard input avasaram ledu.

Example:

```
java Main Harika 20 CSE
```

Values:

args[0] → Harika

args[1] → 20

args[2] → CSE

13. What does args[] represent?

Answer:

args[] anedi **String array**.

Program run chese time lo command line nundi vachina arguments ni store chestundi.

Example:

```
public static void main(String[] args)
```

14. How do you check if command-line arguments are available?

Answer:

args.length use chestam.

Example:

```
if(args.length > 0)
{
    System.out.println(args[0]);
}
else
{
    System.out.println("No arguments provided");
}
```

15. When would you choose Scanner over BufferedReader?

Answer:

Scanner use chestam:

- Beginner programs
- Student practice
- Small applications

- Different data types direct ga read cheyyali anukunte

BufferedReader use chestam:

- Large applications
- High-performance programs
- Huge input data
- Competitive programming (sometimes)